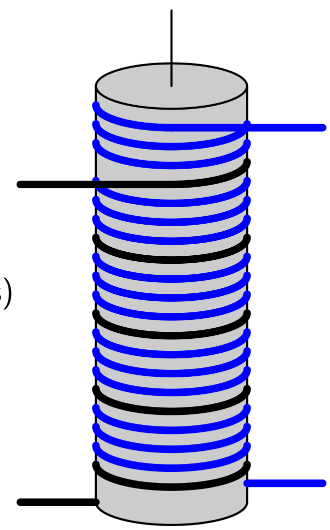
PHYS 3361: Electricity & Magnetism 2
Homework Assignment 3
Due: Friday, Feb 20, 2026 at 5:00PM
Problems marked with **[*]** are potential sources for quiz questions.

Problem 1. Two coils are wrapped around a cylindrical form in such a way that the *same flux passes through every turn of both coils*. (In practice this is achieved by inserting an iron core through the cylinder; this has the effect of concentrating the flux.) The **primary** coil has N_1 turns and the **secondary** has N_2. See the figure below. If the current I in the primary is changing, show that the emf in the secondary is given by

$$\frac{\varepsilon_2}{\varepsilon_1} = \frac{N_2}{N_1},$$

where ε_1 is the (back) emf of the primary. This is a primitive **transformer** - a device for raising or lowering the emf of an alternating current source. By choosing the appropriate number of turns, any desired secondary emf can be obtained.



Problem 2. [*] A transformer (see previous problem) takes an input AC voltage amplitude V_1, and delivers an output voltage of amplitude V_2, which is determined by the turns ratio ($V_2/V_1 = N_2/N_1$). If $N_2 > N_1$, the output voltage is greater than the input voltage. Why doesn't this violate conservation of energy? [*Answer:* Power is the product of voltage and current; if the voltage goes *up*, the current must come *down*. The purpose of this problem is to see exactly how this works out, in a simplified model.]

(a) In an ideal transformer, the same flux passes through all turns of the prmary and of the secondary. Show that in this case $M^2 = L_1 L_2$, where M is the mutual inductance of the coils, and L_1, L_2 are their individual self-inductances.

(b) Suppose the primary is driven with AC voltage $V_{in} = V_1 \cos(\omega t)$, and the secondary is connected to a resistor R. Show that the two currents satisfy the relations

$$L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt} = V_1 \cos(\omega t),$$

and

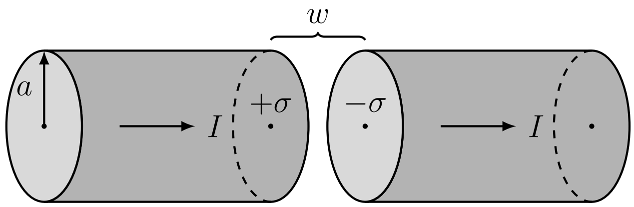
$$L_2 \frac{dI_2}{dt} + M \frac{dI_1}{dt} = -I_2 R.$$

(c) Using the result of part (a), solve these equations for $I_1(t)$ and $I_2(t)$. (Assume I_1 has no DC component.)

(d) Show that the output voltage ($V_{out} = I_2 R$) divided by the input voltage (V_{in}) is equal to the turns ratio: $V_{out}/V_{in} = N_2/N_1$.

(e) Calculate the input power ($P_{in} = V_{in} I_1$) and the output power ($P_{out} = V_{out} I_2$), and show that their averages over a full cycle are equal.

Problem 3. A fat wire, radius a, carries a current I, uniformly distributed over its cross section. A narrow gap in the wire, of width $w \ll a$, forms a parallel-plate capacitor, as illustrated below. Find the magnetic field in the gap, at a distance $s < a$ from the axis.



Problem 4. [*] Consider the charging capacitor in the previous problem.

(a) Find the electric and magnetic fields in the gap, as functions of the distance s from the axis and the time t. (Assume that the charge is zero at $t = 0$.)

(b) Find the energy density u_{em} and the Poynting vector $\boldsymbol{S}$ in the gap. Note especially the *direction* of $\boldsymbol{S}$. Check that

$$\frac{\partial u}{\partial t} = -\boldsymbol{\nabla} \cdot \boldsymbol{S}$$

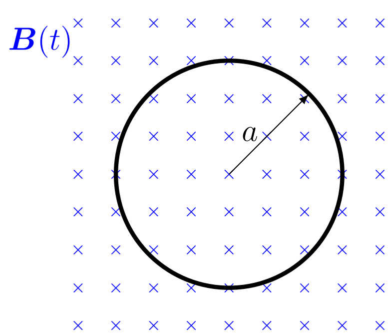
is satisfied.

(c) Determine the total energy in the gap, as a function of time. Calculate the total power flowing into the gap, by integrating the Poynting vector over the appropriate surface. Check that the power input is equal to the rate of increase of energy in the gap

$$\frac{dW}{dt} = -\frac{d}{dt}\int_{\mathcal{V}} \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) d\tau - \frac{1}{\mu_0}\oint_{\mathcal{S}} (\boldsymbol{E} \times \boldsymbol{B}) \cdot d\boldsymbol{a}.$$

Note: $W = 0$ in the above because there is no charge in the gap.

Problem 5. This problem is dedicated to John. When he held the aluminum ring over the iron core of the solenoid he commented on the vibration (and heat!) he felt. Consider a circular conducting ring of radius a in a time-changing magnetic field as illustrated below.



(a) Assuming that the magnetic field has a time dependence of $B(t) = B_0 \sin(\omega t)$, compute the emf induced in the ring if John is holding the ring steady. You should find,

$$\varepsilon(t) = B_0 \pi a^2 \omega \cos(\omega t).$$

(b) Assume that the ring has resistance R and inductance L. Use Kirchoff's loop rule to write down an equation for the voltages around the ring. You should find:

$$L\frac{dI}{dt} + R_{\text{tot}} I = B_0 \pi a^2 \omega \cos(\omega t).$$

(c) Assuming John has held the ring long enough for transients to die out; find the steady state current, $I(t)$, in the ring. You should find:

$$I(t) = \frac{B_0 \pi a^2 \omega}{\sqrt{R^2 + (\omega L)^2}} \left(R\cos(\omega t) + \omega L \sin(\omega t)\right) = -I_0 \cos(\omega t - \phi),$$

with $I_0 = \omega B_0 A / \sqrt{R^2 + (\omega L)^2}$ and $\phi = \tan^{-1}(\omega L/R)$.

(d) Compute the power dissipated in the ring and find its time average over a cycle and show that it is $\langle P \rangle = R I_0^2/2$.

(e) Assuming that all this time-averaged power goes into heating the ring, which has a mass, m, and a specific heat capacity, c_p, determine an equation for the temperature of the ring as a function of time, $T(t)$. You should find

$$T(t) = T_{room} + \frac{1}{2}\frac{R(\omega B_0 A)^2}{(R^2 + (\omega L)^2) m c_p} t.$$

(f) Return to $I(t)$ and compute $d\boldsymbol{F} = I(t) d\boldsymbol{\ell} \times \boldsymbol{B}(t)$, where $d\boldsymbol{\ell}$ points in the $\hat{\boldsymbol{\phi}}$ direction around the ring. Integrate this around the ring to find the net force on the ring. You should find

$$F(t) = -\pi a I_0 B_0 \sin\phi - \pi a I_0 B_0 \sin(2\omega t - \phi).$$

What direction does this point? What do you notice about this expression?

(g) Let's put some numbers to these calculated quantities. Reasonable assumptions about the ring and solenoid magnetic field are: $m = 24$-g, $a = 3.0$-cm, $h = 1.5$-cm, $t = 3$-mm, $B_0 = 0.1$-T.

(i) Compute the ring's electrical resistance. Look up the value of the resistivity of aluminum and treat the ring as a wire with cross-section given by the height and thickness of the ring and length equal to the circumference of the ring. You should find a resistance value $\approx 70\mu\Omega$.

(ii) Compute the ring's inductance using the formula,

$$L = \mu_0 a \left[\ln\left(\frac{8a}{r_{eq}}\right) - 2\right],$$

where $r_{eq} = \sqrt{th/\pi}$ is an "equivalent" radius of the ring's rectangular conducting cross-sectional area. You should find a value for the ring's inductance of ≈ 50-nH. Where this formula for a single loop's inductance comes from is quite involved so we will just take it on faith for now. (Unless you want to work that problem then let me know...)

(iii) Compute ωL and compare to R. Can one be ignored or are they both about the same size?

(h) Compute I_0. What do you think about that number?

(i) Plot the temperature of the ring, $T(t)$, vs time. Assume the ring starts at room temperature. Generally speaking, something is "too hot to hold" for humans in the 45° - 50° C range. Is our model accurate (within a factor of 2 or so) in estimating when John exclaimed, "It's getting hot!"?

(j) Compute the magnitude of the constant force on the ring. What is this force trying to do to the ring?

(k) Compute the amplitude of the time-varying force on the ring. The vibration that John felt is probably partly this (at 2ω) and a bit of vibration due to a time-varying torque (at ω) due to misalignment of the ring with the magnetic field as well as its warped shape.